

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/221543598>

# Spanning Closed Trail and Hamiltonian Cycle in Grid Graphs.

Conference Paper · December 1995

DOI: 10.1007/BFb0015440 · Source: DBLP

---

CITATIONS

6

---

READS

91

2 authors:



**Hwan-Gue Cho**

Pusan National University

267 PUBLICATIONS 2,045 CITATIONS

SEE PROFILE



**A. Zelikovsky**

Georgia State University

323 PUBLICATIONS 7,024 CITATIONS

SEE PROFILE

# Spanning Closed Trail and Hamiltonian Cycle in Grid Graphs <sup>\*</sup>

Hwan-Gue, Cho<sup>1</sup> and Alexander Zelikovsky<sup>2</sup>

<sup>1</sup> Dept. of Computer Science, Pusan National University,  
Pusan 609-735, Korea, hgcho@hyowon.cc.pusan.ac.kr

<sup>2</sup> Institute of Mathematics, Academiei 5  
Kishinev, Moldova, 17azz@mathem.moldova.su

**Abstract.** In this paper we study a trail routing and a hamiltonian cycle in a class of grid graphs, *polycube* and *polymino*. A *Spanning closed trail* is an eulerian subgraph containing all vertices of a given graph. For general grid graphs we prove that the problem of finding that trail is *NP*-complete and for a wide subclass of grid graphs, called *polymino*, we give an optimal algorithm if it exists. For polycube graphs we prove that every polycube has a spanning closed trail. Finally we show that a graph product  $G$  to a simple path with length  $n$ ,  $G \times P_n$ , is hamiltonian for all  $n \geq 2$ , if  $G$  is a polymino with a perfect matching.

## 1 Introduction

Spanning closed trail(for brevity trail) can be considered as one relaxation between two cycles, hamiltonian and eulerian cycle. While hamiltonian routing is to visit all vertices(at most once) with some edges(at most once) and eulerian routing is to visit all edges(at most once) with all vertices(at least once), this trail routing is to visit all vertices(at least once) with some edges(at most once).

We know the spanning trail can be applied in practice such as *one-way street problem* and *optimal page retrieval problem*[10, 8]. It is obvious that if a graph  $G$  has a trail,  $G$  is *supereulerian* graph[4]. There are interesting characterizations on supereulerian graphs[1, 2, 3, 4]. They gave some sufficient conditions for supereulerian graphs, but did not show any algorithms to find a trail.

## 2 Spanning Closed Trail in Polyminos

In the following a *path* means an alternating sequence of vertices and edges and a *trail* is a path with all distinct edges[5]. The planar *rectangular grid* consists of vertices  $\{(x, y) | x, y \in Z\}$  and edges  $\{(u, v) | u, v \in V, |u - v| = 1\}$ . A *grid graph*  $R = (V, E)$  is a finite subgraph of the planar rectangular grid [7]. A bounded face  $F$  of  $R$  with exactly four vertices on its boundary is called a *cell*, all other

---

<sup>\*</sup> Parts of this work were done while authors visited in Max-Planck-Institut für Informatik, Saarbrücken, Germany.

bounded faces are called *holes*. A grid graph is a *polymino* if every edge is one of boundary edges of a unit cell. A *simple polymino* is a 2-connected grid graph without holes. This section is devoted to the following.

**Problem 1. The Trail Problem:** Given a grid graph  $R$ , find a trail for  $R$  if it exists.

Note that if a hamiltonian cycle exists, then it is the shortest trail. The Hamilton circuit problem is  $NP$ -complete for grid graphs, but it is still unknown whether this problem is  $NP$ -complete for simple polyminoes (see [6]).

We show that the Trail Problem can be solved in linear time for simple polyminoes and so called *cell connected* polyminoes with holes. From the other side we prove that the Trail Problem for arbitrary polyminoes is  $NP$ -complete.

## 2.1 Trails in Simple Polyminos

Let  $R$  be a simple polymino bounded by a simple polygon  $P$  and let  $T$  be its trail. Since  $T$  is a planar eulerian graph, the dual graph for  $T$  is bipartite. Color all faces of  $T$  into two colors black and white, such that the unbounded face is white. Any cell obtains the color of the face it belongs to. There is one to one correspondence between trails of  $R$  and *proper* colorings of cells such that

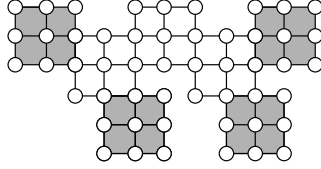
- (1) the cells of the unbounded face are white;
- (2) the edges of black cells form a connected subgraph;
- (3) each vertex belongs to at least one black and one white cell.

We need the following definitions. The *cell graph* of  $R$  is a graph on cells of  $R$  such that two cells are adjacent if they have an edge in common. So the cell graph is a dual graph for  $R$  without a vertex corresponding to the unbounded face.  $R$  is called *cell-connected* if its cell graph is connected. A *corner* of a polymino is a vertex of degree 2. An edge  $e$  of  $R$  is a *neck* if  $e \notin P$  and the both ends of  $e$  belong to  $P$ . *Inner* vertices of  $R$  do not belong to its boundary  $P$ . An inner vertex is an *eye* if all its four neighbors belong to the boundary. A *block* is a polymino bounded by a square with a side of size 2. A *dragon* is a simple polymino built from blocks such that any two blocks are disjoint or may have in common exactly one edge (Fig. 1). Inner vertices of a dragon are *eyes*. A *head* of a polymino  $R$  is a block with one edge in common with the rest of  $R$ . Heads are darkened in Fig. 1. The following lemma is obvious.

**Lemma 2.** *Any dragon except a block has at least two heads.*

**Lemma 3.** *Dragons have no trails.*

*Proof.* In a proper coloring all corner cells of a head should be black and the rest cells of a head should be white. Therefore if we remove a head from a dragon, then the rest dragon should have a proper coloring also (Fig. 2 (a) ). Removing heads one by one leads to a block which has no proper coloring, since all its cells should be black and eye cannot belong to a white cell.



**Fig. 1.** A dragon with dark heads.

The main result of this section says that the converse statement is also true.

**Theorem 4.** *A simple polymino has a closed spanning trail if and only if it is not a dragon.*

To prove Theorem 4 we show the way how to find a trail if it exists.

**Theorem 5.** *A spanning closed trail for a simple polymino, if it exists, can be found in linear time.*

*Proof.* First we consider the case of a polymino  $R$  without eyes and necks. We show that the *chess* coloring of  $R$  defined below is proper. Initially all cells of the grid are white. We color black a cell  $c$  of  $R$  if (i)  $c$  contains a corner of  $R$  and/or (ii) the sum of coordinates of the left low corner of  $c$  is even.

The chess coloring satisfies the condition (2) since  $R$  does not contain necks and the cell graph of  $R$  is connected. Note that each vertex belongs to at least one black cell. Any boundary vertex obviously belong to at least one white cell of the unbounded face. Any inner vertex has at least one inner neighbor. These two inner vertices belong to two cells with an inner edge in common. One of these two cells is white. Thus we proved that the chess coloring is proper.

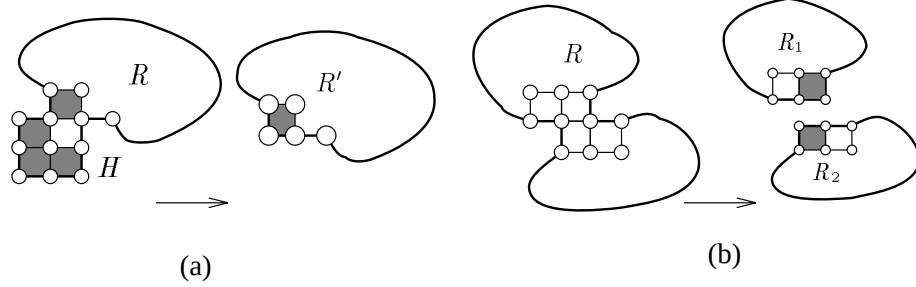
Now assume that  $R$  is not a dragon. Below we present several linear-time procedures which transform  $R$  to a set of polyminoes without eyes and necks.  $Cut_{head}$  replaces a head  $H$  of  $R$  by one edge (Fig. 2 (a)). Walking along the boundary polygon  $P$  we apply  $Cut_{head}$  to the head met first. If as a result a new head appears, then cut this new one and so on. If a new head does not appear, then continue our walk along  $P$  until we return to the initial vertex.

$Cut_{neck}$  (Fig. 2 (b)) gives two simple polyminoes from the given polyminoes such that both contain the neck edge and call  $Cut_{head}$ .

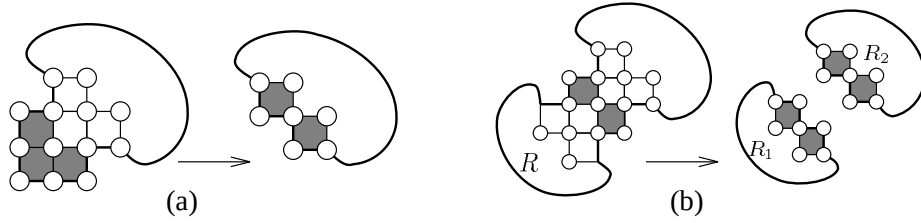
Still there are two cases when  $R$  contains eyes. A *bump* is a union of three corners slightly different from a head (Fig. 3 (a)). A *tumor* looks like a bump inside the polymino).  $Cut_{bump}$  and  $Cut_{tumor}$  remove a block adding two or four edges to  $P$ , respectively (Fig. 3(a) and 3(b)).

It is not difficult to see that any proper coloring of polyminoes obtained after application of procedures above can be extended to a proper coloring of the initial polymino. Thus we apply to  $R$   $Cut_{heads}$ ,  $Cut_{neck}$  for each neck every time removing heads appeared, and finally  $Cut_{bumps}$  and  $Cut_{tumors}$  for each bump and tumor, respectively.

Simple polyminoes resulted do not contain eyes and necks, so the chess coloring is proper for them. Then we can easily obtain a proper coloring of the initial simple polymino running cut procedures in the reverse order.



**Fig. 2.** Procedures for (a)  $Cut_{head}$  and  $Cut_{neck}$ .



**Fig. 3.** Procedures for (a)  $Cut_{bump}$  and (b)  $Cut_{tumor}$ .

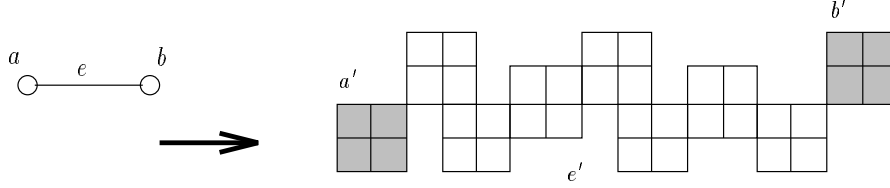
## 2.2 The Trail Problem in Polymino with Holes

In this subsection we will prove a negative result on Trail Problem (TP) for grid graphs with holes. This result for TP is somehow stronger than the negative result for the Hamilton circuit problem since it holds not for arbitrary grid graphs but for polyminoes. We also show that the Trail Problem for cell-connected polyminoes can be solved in linear time.

**Theorem 6.** *The Trail Problem in polymino with holes is NP-complete.*

*Proof.* Our proof is similar to the proof in [6]. We transform the Hamilton Circuit Problem (HCP) in planar cubic maps to Trail Problem. For each vertex of a cubic

map we assign points of a block. For each edge of a cubic map  $M$  we replace it by a dragon with two heads, i.e. a *chain-dragon*. Let an edge  $e$  be incident to a vertex  $a$  in a cubic map. Then the corresponding block  $a'$  and chain-dragon  $e'$  have a common corner (Fig. 4).



**Fig. 4.** A polymino image of a cubic map edge.

**Lemma 7.** *Any chain-dragon has a Hamiltonian path  $P$  which connects any pair  $a$  and  $b$  of corners from different heads. Moreover  $P$  may go along arbitrary two edges incident to  $a$  and  $b$ , respectively.*

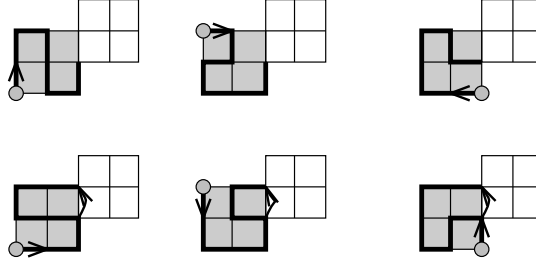
*Proof.* A head has three corners and each corner has two incident edges. So it is enough to consider only 6 cases. Fig. 5 illustrates how to span a dark head with a hamiltonian path such that after cutting a dark head we need only continue our path in the same manner. Hamiltonian paths of the head appeared after cutting the dark head should begin from the left lower corner. In the cases illustrated by the lower (Fig. 5), we subtract from such a path an edge marked by directed curve.

Assume that  $M$  has a hamiltonian cycle. In the corresponding polymino  $M'$  we can span all images of vertices and Hamiltonian edges with the corresponding cycle. The rest of vertices, i.e vertices of chain-dragons, we can span with the corresponding trails, since we can omit one corner of such chain-dragon.

Suppose we have a trail  $T$  for the polymino  $M'$ . This trail  $T$  cannot span one chain-dragon unless it contains a hamiltonian path between the corresponding two corners of its heads. The chain-dragons which are spanned by paths form some cycle. This cycle should correspond to some hamiltonian cycle  $H$  in the original map  $M$ , since all other chain-dragons cannot connect images of vertices and  $T$  should be disconnected if  $H$  does not cover all points of  $M$ .

Still we can solve the Trail Problem in the *tree* polymino in which any 2-connected component is cell-connected.

**Theorem 8.** *The Trail Problem for tree polyminoes can be solved in linear time. A tree polymino has a trail if and only if it does not contain a dragon as its 2-connected component.*



**Fig. 5.** A dark head spanned with a thick hamiltonian path.

*Proof.* Let  $R$  be a cell-connected polymino with at least one hole. We apply the algorithm for simple polyminoes to  $R$ . A dragon may appear only while application of the procedure *cut necks*. Let  $R'$  be a dragon obtained after cutting a neck from  $R$ .  $R'$  can have only two heads and  $R$  should be a chain-dragon. By Lemma 7 we can span all points of  $R'$  by a hamiltonian path  $P$  connecting corners of two heads of  $R'$  which correspond to two vertices of the neck of  $R$ . Moreover we may lead  $P$  through two edges which correspond to the neck of  $R$ . This hamiltonian path corresponds to some hamiltonian cycle of  $R$  with a doubled neck.

In the following section we will consider Trail Problem in 3-dimensional grids, called *polycube* graph.

### 3 Spanning closed trail in polycube graphs

In this section we give an optimal algorithm for the trail in a class of *cubic grid* graphs. The cubic grid graph consists of vertices  $\{ (x, y, z) | x, y, z \in \mathbb{Z} \}$  and edges  $\{ (p, q) | p, q \in V \text{ and } |p - q| = 1 \}$ . A *polycube graph*  $R = (V, E)$  is a finite subgraph of the cubic grid graph. If a polycube does not have any empty space in its internal body, then we call it *simple polycube*.

Previous sections proved that every simple polymino has a trail if it is not a dragon polymino. This section shows that every simple polycube has a spanning closed trail, even though a polycube graph has a *dragon polymino* in one of its layers. In the following  $i$ -th layer of a polycube  $G$  denotes the plane polymino subgraph which is horizontally embedded in  $G$  with height  $i$ .

**Lemma 9.** *Let  $G_p$  be a polycube. If  $(i + 1)$ -th layer of  $G_p$  is not a dragon and there is a closed trail which spans upto  $i$ -th layer from the bottom, then there is a trail spanning upto  $(i + 1)$ -th layer.*

*Proof.* Let  $t_{i+1}$  be a trail in the  $(i + 1)$ -th layer.  $t_{i+1}$  can be found easily since  $(i + 1)$ -th layer is not a dragon. And let  $T_i$  be a closed trail upto  $i$ -th layer from

0-th layer. Since  $G_p$  is a polycube structure, at least one unit polycube is shared by every two adjacent layers. So there is an edge  $(x, y)$  in  $(i + 1)$ -th layer which has a corresponding edge  $(x', y')$  in  $i$ -th layer such that where  $(x, x'), (y, y') \in E(G_p)$ .

Suppose that  $(x, y)$  is included in  $t_{i+1}$ . If  $(x', y')$  is not included in  $T_i$ , we know that  $t_{i+1} + T_i - (x, y) + (x', y') + (x, x') + (y, y')$  gives a bigger closed trail for upto  $(i + 1)$ -th layer from the base layer. Otherwise if  $(x', y')$  is included in  $T_i$ , then  $t_{i+1} + T_i - (x, y) - (x', y') + (x, x') + (y, y')$  also gives another trail upto  $(i + 1)$ -th layer.

If  $(x, y)$  is not included in  $t_{i+1}$ , we can get a closed trail similarly.

**Lemma 10.** *If  $D$  is a dragon polymino, then there is a spanning trail(not closed) which starts with  $x$  and ends at  $y$ , for every edge  $(x, y)$  of  $D$ .*

*Proof.* Let  $[x, y]$ -trail denote a spanning trail (not closed) which starts with  $x$  and ends at  $y$  in a layer. We construct  $[x, y]$ -trail by a constructive procedure on the number of blocks. In a single block  $B$ , it is easy to see that there is  $[x, y]$ -trail for every edge  $(x, y)$  of  $B$ .

We show how to find the  $[x, y]$ -trail for a given  $(x, y)$  edge in  $D$ . Let  $B_1$  be a single block of  $D$  which contains  $(x, y)$ . The whole  $D$  can be constructed by attaching blocks to  $B_1$  one by one. Let  $\{B_i\}$  denote a sequence of blocks attached to rebuild  $D$  from  $B_1$ .

Now we consider the step of attaching  $B_2$  to one of boundary edges in  $B_1$ . Let  $(p, q)$  be the neck edge shared by two heads  $B_1$  and  $B_2$ . Since there is  $[p, q]$ -trail in  $B_2$  for any boundary edge  $(p, q)$ , we can merge these two trails,  $[x, y]$ -trail of  $B_1$  and  $[p, q]$ -trail of  $B_2$  into a bigger  $[x, y]$ -trail which spans two headed dragon  $B_1 + B_2$ . By this constructive procedure we can make  $[x, y]$ -trail for  $D$  by attaching  $B_i$  to a neck edge of the intermediate dragon made of upto  $B_{i-1}$  blocks added.

**Lemma 11.** *Let  $G_P$  be a polycube. Suppose that its  $(i + 1)$ -th layer is a dragon. If there is a closed trail which spans upto  $i$ -th layer, then we can make a closed trail upto  $(i + 1)$ -th layer.*

*Proof.* According to the proof of Lemma 9, we know that there exists at least one edge  $(x, y)$  in  $(i + 1)$ -th layer such that  $(x, x'), (y, y') \in E(G_p)$  and  $(x', y')$  is an edge of  $i$ -th layer. Then we can find  $[x, y]$ -trail in  $(i + 1)$ -th layer by Lemma 10. Let  $t_{i+1}$  be  $[x, y]$ -trail of  $(i + 1)$ -th layer and  $T_i$  be the closed trail upto  $i$ -th layer. If  $(x, y)$  is not included in  $T_i$ , then  $t_{i+1} + T_i + (x, x') + (y, y') + (x', y')$  gives a closed trail spanning upto  $(i + 1)$ -th layer from base layer. Otherwise if  $(x, y)$  is included in  $T_i$ , then by the similar way of proof in Lemma 9, we can construct another trail for upto  $(i + 1)$ -layer.

**Lemma 12.** *Let  $G_p$  be a polycube graph and both  $i$ -th and  $(i+1)$ -th layer are dragons. Then there is a closed trail in both layers.*



*Proof.* There are a pair of corresponding edges such that  $(x, y)$  and  $(x', y')$  is an edge of  $i$ -th and  $(i + 1)$ -th layer, respectively and  $(x, x'), (y, y') \in E(G_p)$  by Lemma 9. By Lemma 10 we can find  $[x, y]$ -trail in  $i$ -th layer and  $[x', y']$ -trail in  $(i + 1)$ -th layer. Then adding two additional edges such that  $[x, y]$ -trail +  $[x', y']$ -trail +  $(x, x') + (y, y')$  gives a closed trail which spans both dragon layers.

The following theorem gives the main result of this section.

**Theorem 13.** *Every simple polycube has a spanning closed trail. And we can find it in linear time.*

*Proof.* In order to find a trail of a simple polycube, we construct it by merging trails of each layer one by one from the base layer. Combining all above lemmas in this section completes this theorem.

For merging trails, firstly we have to find edge  $(x, y)$  in  $(i + 1)$ -th layer with the corresponding edge  $(x', y')$  in  $i$ -th layer. It is easy to find such an edge by looking all edges in one layer. This work can be done within  $O(n_i)$  steps, where  $n_i$  is the number of edges in  $i$ -th layer.

To merge two trails into one trail, it is enough to consider at most four edges by Lemma 9 and 10. Therefore the total number of edges examined to find a closed trail for a polycube  $G_p$  is  $O(|E(G_p)|)$ .

## 4 Hamiltonian cycle in polycube graphs

Till now there are a little results on the hamiltoncity of grid graphs. They gave a necessary and sufficient condition for the existence of the hamiltonian path in rectangular grids [6]. Also they have shown that the hamiltonian problem for general grids is  $NP$ -complete. But it should be noted that their grid graph is different from our polymino graphs in that our polymino model does not allow any degenerated faces or holes.

This section introduces  $n$ -layer graph as one extension of planar grids. For a given planar graph  $G$ , the  $n$ -layer graph of  $G$  is defined as the graph product  $G \times P_n$  denoted as  $(G \times n)$ , where  $P_n$  is a linear path with length  $n$ .

A polymino is said to have an even(odd) parity, if it is (not) a bipartite graph with even partition of vertices. We know that if a polymino has an even parity then it has a perfect matching.

Though little was known about the properties of polymino, we conjecture that deciding hamiltoncity of a simple polymino is  $NP$ -complete even though that graph has an even parity without holes. The main result of this section is stated.

**Lemma 14.** *If  $G$  is a simple polymino,  $(G \times 2n)$  is hamiltonian for  $n \geq 1$ .*

*Proof.* It is trivial that a rectangular grid with a side of even length is hamiltonian. So we decompose  $G$  into vertical layers, find hamiltonian cycles for each layer and merge all hamiltonian cycles into one.

**Theorem 15.** *If  $G$  is a simple polymino with a perfect matching  $M$ , then  $n$ -layer  $(G \times n)$  is hamiltonian for  $n \geq 2$ .*

*Proof.* It is enough to prove  $(G \times 3)$  is hamiltonian. At first we find a perfect matching  $M$ . Then we get Depth First Search tree with matching  $M$ ,  $DFS_M$ , which is a spanning tree including all  $M$ . This  $DFS_M$  searching is different to the original  $DFS$  in that  $DFS_M$  prefers the matched edge in searching procedure.

Assume  $DFS_M$  procedure just has visited vertex  $p$  and there are several unvisited edges incident to  $p$ . If there is an edge  $(p, q) \in M$  and it is not still included in the current search tree, then we include  $(p, q)$  rather than visiting other unmatched edges. If there is no matched edges incident to  $p$ , then we take an arbitrary unmatched and unvisited edge.

Let  $S_M$  be a spanning tree obtained by  $DFS_M$  searching. It should be noted that  $S_M$  tree has no vertex whose degree is greater than 4 since  $G$  is a 2-connected polymino. If a vertex  $x$  is of degree 4 in  $S_M$ , then one of its adjacent vertex should have been visited previously along the path from the other adjacent vertices of  $p$  in  $S_M$  due to the 2-connectivity of  $G$ .

Fig. 6 shows that  $x$  and  $y$  must be in the same 2-connected component, so there is a path connecting  $x$  and  $y$ . So  $p$  of  $S_M$  can not be of degree 4 by the property of  $DFS_M$ . In Fig.6(left top) thick edges (solid thin edges) denote  $M$  ( $S_M$  edges not included in  $M$ ).

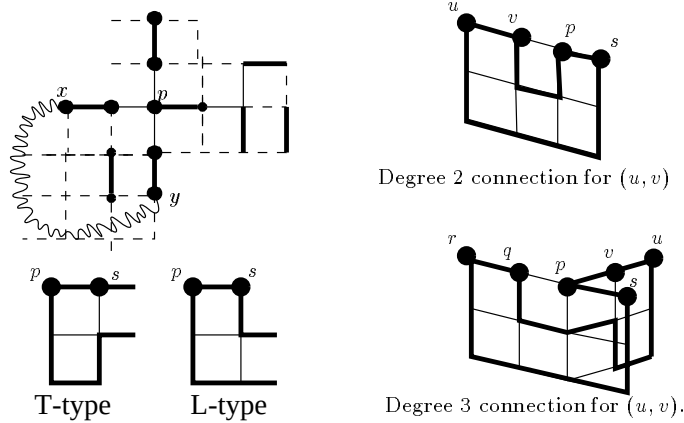
Now we construct a hamiltonian cycle with  $S_M$ . It is obvious that if  $|M| = 2$  then  $(S_M \times 3)$  has a hamiltonian cycle. As an induction base we assume that if  $|M| \leq i$  then  $(S_M \times 3)$  is hamiltonian. Suppose that  $|M| = i$  and  $S_M$  is given, then we make another spanning tree  $S'_M$  by deleting a leaf edge  $m = (u, v)$  of  $S_M$ . Let  $u$  be a leaf vertex in  $S_M$ , then it is obvious that  $m = (u, v)$  should be a matched edge.

By induction hypothesis we can construct a hamiltonian cycle in  $(S'_M \times 3)$ . Now we restore  $m$  to  $S'_M$  to make a bigger hamiltonian cycle for  $(S_M \times 3)$ . In order to restore  $(u, v)$  with preserving hamiltonicity, we prepare  $L$  and  $T$ -type components. Since the degree of vertex  $v$  of  $m$  is 2 or 3 in  $S_M$ , we can easily select  $L$  or  $T$ -type path to extend the hamiltonian cycle of  $S'_M$  in every leaf edge restoring step. Illustrations of degree 2, 3 connections in Fig.6 proves its validity and correctness easily. This completes the proof.

**Corollary 16.** *If  $G$  is a 2-connected planar graph with a perfect matching, then  $(G \times n)$  is hamiltonian.*

## 5 Concluding Remarks

In this paper we proved that all polymino graphs except dragons are supereulerian and every simple polycube is supereulerian. And our proof procedure implies an optimal linear time algorithm. And we gave one class of hamiltonian graph such as  $(G \times n)$  where  $G$  is 2-connected planar graph with a perfect matching. Now we are trying to find the longest trail in a simple polymino. We hope to see other applications for trails and a hamiltonian routing in a polycube.



**Fig. 6.**  $(p, s)$ ,  $(r, q)$  are the matched edges.  $m = (u, v)$  is restored to  $S'_M$

## References

1. P. A. Catlin, *A reduction method to Find Spanning Eulerian Subgraphs*, J. Graph theory, Vol.12, No. 1, pp.29-44, 1988.
2. P. A. Catlin, *Spanning trails*, J. Graph Theory, vol.11, No.2, pp.161-167, 1987.
3. P. A. Catlin, *Spanning Eulerian subgraphs and matching*, Discrete Math., 76, pp. 95-116, 1989
4. Z. H. Chen, *Spanning closed trails in graphs*. Discrete Math. 117, pp. 57-71, 1993.
5. F. Harary, *Graph Theory*, Addison-Wesley Pub., 1972.
6. A. Itai, C. H. Papadimitriou and J.L. Szwarcfiter, *Hamiltonian Paths in Grid Graphs.*, SIAM, Comp., Vol., 11, No.4, Nov., pp.676-686, 1982.
7. M. Kaufmann and K. Mehlhorn, *A linear-time algorithm for the Homotopic routing in grid graph* SIAM J. Comput. vol. 23, No. 2. pp. 227-246, Apr. 1994.
8. T. H. Lai and L. S. Wei, *The edge hamiltonian path problem is NP-complete for bipartite graphs* Infom. Process. Lett., 46, pp. 21-26, 1993.
9. C.H. Papadimitriou and U.V. Vazirani, *On two geometric problems related to the traveling salesman problem*, J. of Algorithms, vol.5, pp.231-246, 1984.
10. F. S. Roberts, *Graph Theory and Its applications to Problems of Society*, CBMS-NSF Regional Conference Series in Applied Mathematics, Vol. 29, SIAM, 3rd, pr. 1993.